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**CS 5588: Homework 1**

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## 1. Selections of AI tools, programming language, and xUnit.

|                      | Example   | Your Selection |
|----------------------|-----------|----------------|
| AI Tool & Version    | ChatGPT 5 | Claude Opus 5  |
| Programming Language | Java      | Java           |
| xUnit                | JUnit     | JUnit          |

## 2. Adjusted Average

### 2.1

#### 1. Prompts/Response

##### a. Prompt:

- Create a Java folder. Open that folder here and write a method or function, `adjustedAverage`, that calculates the average of a given list of integer values, excluding one occurrence of the maximum and minimum values. Provide at least 10 xUnit tests for `adjustedAverage` and fix all errors and test failures.

##### b. Response:

- Opened the CS 5588 HW1 folder and created:
  - `AdjustedAverage.java`
  - `AdjustedAverageTest.java`
  - `pom.xml`
- The implementation removes one minimum and one maximum, with 13 JUnit 5 tests covering normal cases and invalid input. Editor error checks are clean.

#### 2. Screenshots of test execution results

```
joshexe@mac CS 5588 HW1 % mvn test
[INFO] Scanning for projects...
[INFO]
[INFO] -----< edu.cs5588:cs5588-hw1 >-----
[INFO] Building cs5588-hw1 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ cs5588-hw1 ---
[INFO] skip non existing resourceDirectory /Users/joshexe/CS 5588 HW1/src/main/resources
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ cs5588-hw1 ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ cs5588-hw1 ---
[INFO] skip non existing resourceDirectory /Users/joshexe/CS 5588 HW1/src/test/resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ cs5588-hw1 ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ cs5588-hw1 ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running AdjustedAverageTest
[INFO] Tests run: 13, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.038 s -- in AdjustedAverageTest
[INFO] Results:
[INFO]
[INFO] Tests run: 13, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 0.796 s
[INFO] Finished at: 2026-09-10T17:52:09-05:00
[INFO] -----
```

### 3. Discussion of AI-generated errors/fixes

- a. The generated implementation was correct, but the generated tests only covered ordinary lists. I added cases for duplicate minimums and maximums, all-equal values, mixed signs, order independence and null elements, since an off-by-one in the removal logic would show up in exactly those cases

### 4. Source Code: <https://github.com/JUSH334/CS-5588-Homework-1.git>

## 2.2

### 1. Prompts/Responses

#### a. Prompt:

- i. Determine whether the `adjustedAverage` implementation suffers from integer overflow. If overflow is impossible, provide a justification. Otherwise, modify the program to prevent overflow, generate at least five xUnit tests that specifically verify overflow handling, and fix all errors and test failures.

#### b. Response:

- i. The `'adjustedAverage'` implementation does not suffer from integer overflow. Although the input values are `'Integer's`, the method stores their sum in a `'long'`, which can hold a much larger range of values. A Java list cannot contain more than `'Integer.MAX_VALUE'` elements, and each element is at most `'Integer.MAX_VALUE'` or `'Integer.MIN_VALUE'`. Therefore, even the largest possible sum remains within the range of a `'long'`. The minimum and maximum values are also subtracted using `'long'` arithmetic, so the calculation remains safe. Because overflow is impossible under these constraints, no changes to the implementation or additional overflow-specific tests are necessary.

#### c. Prompt:

- i. Can you generate at least five xUnit tests that specifically verify overflow handling, and fix all errors and test failures to further make sure?

#### d. Response:

- i. Added five overflow-boundary tests to `AdjustedAverageTest.java`:
  1. All `Integer.MAX_VALUE` values
  2. All `Integer.MIN_VALUE` values
  3. Opposite integer extremes
  4. Two pairs of extreme values
  5. Extreme values after removing outliers

The implementation was unchanged because it already uses a long accumulator. All 18 tests passed.

## 2. Screenshot of test execution results

```
INFO] -----
joshexe@Joshuas-MacBook-Air CS 5588 HW1 % mvn test -Dtest=AdjustedAverageTest
INFO] Scanning for projects...
INFO] -----< edu.cs5588:cs5588-hw1 >-----
INFO] Building cs5588-hw1 1.0-SNAPSHOT
INFO] from pom.xml
INFO] -----[ jar ]-----
INFO]
INFO] --- resources:3.4.0:resources (default-resources) @ cs5588-hw1 ---
INFO] skip non existing resourceDirectory /Users/joshexe/CS 5588 HW1/src/main/resources
INFO]
INFO] --- compiler:3.15.0:compile (default-compile) @ cs5588-hw1 ---
INFO] Nothing to compile - all classes are up to date.
INFO]
INFO] --- resources:3.4.0:testResources (default-testResources) @ cs5588-hw1 ---
INFO] skip non existing resourceDirectory /Users/joshexe/CS 5588 HW1/src/test/resources
INFO]
INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ cs5588-hw1 ---
INFO] Nothing to compile - all classes are up to date.
INFO]
INFO] --- surefire:3.2.5:test (default-test) @ cs5588-hw1 ---
INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
INFO]
INFO] -----
INFO] T E S T S
INFO] -----
INFO] Running AdjustedAverageTest
INFO] Tests run: 18, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.043 s -- in AdjustedAverageTest
INFO]
INFO] Results:
INFO]
INFO] Tests run: 18, Failures: 0, Errors: 0, Skipped: 0
INFO]
INFO] -----
INFO] BUILD SUCCESS
INFO] -----
INFO] Total time: 0.895 s
```

## 3. Discussion of AI-Generated errors/fixes

- a. The AI's answer was correct but untested, so I asked it to prove the claim in code. The five overflow tests all pass against the unchanged implementation: all Integer.MAX\_VALUE, all Integer.MIN\_VALUE, opposite extremes, a 1000-element list of Integer.MAX\_VALUE, and extreme values surrounding moderate ones.

## 4. Source Code: <https://github.com/JUSH334/CS-5588-Homework-1.git>

## 3. Syllable Counter

### 3.1

#### 1. Prompts/Response

##### a. Prompt:

- i. Write a method or function, SyllableCounter, that counts the syllables in the given English word, provide at least five xUnit tests, and fix all errors and test failures.

##### b. Response:

Created SyllableCounter.countSyllables(String word), which counts syllables from spelling alone, with no table of known words.

The count starts as the number of vowel groups in the word, treating y as a vowel, then applies two sets of corrections, as the counter stands after the section 3.2 fixes:

1. **Silent endings are removed before counting.** The final *e* of *whale*, the *-ed* of *jumped*, and the *-es* of *makes* add letters but no syllable. The *-ed* is kept when it follows *t* or *d* (*wanted*, *needed*), and the *-es* is

kept after a hissing sound (*boxes, watches*). A consonant plus *-le* keeps its syllable, as in *table* and *purple*.

2. **Spelling patterns adjust the count by one.** Adjacent vowels that are spoken separately add a syllable — *vi-o-lin, cre-ate, po-em, be-ing, i-de-a* — while the same letter pairs after *c, s, t* or *g* are spoken together and are excluded, so *na-tion, re-gion* and *spe-cial* are unaffected. The *-ely* ending removes one, as in *late-ly*. The result is floored at 1, since every English word has at least one syllable. The *-ia* and *-ely* patterns were added in section 3.2, when the hardcoded word table was removed; they are listed here so the description of the algorithm is complete.

Wrote 38 JUnit 5 tests for the algorithm itself, grouped by the rule each one exercises:

| Rule under test                   | Examples                              | Tests |
|-----------------------------------|---------------------------------------|-------|
| One-syllable words                | book, strength, the                   | 3     |
| One syllable per vowel group      | water, banana, elephant               | 3     |
| Silent final <i>e</i>             | whale, smile, machine                 | 3     |
| Consonant plus <i>-le</i>         | table, purple, people                 | 3     |
| <i>y</i> as a vowel               | sky, style, happy, yellow, family     | 5     |
| Silent <i>-ed</i> and <i>-es</i>  | jumped, makes, clothes                | 3     |
| Spoken <i>-ed</i> and <i>-es</i>  | wanted, needed, boxes, watches        | 4     |
| Adjacent vowels spoken separately | create, poem, being, violin, idea     | 5     |
| Adjacent vowels spoken together   | nation, region, special, during, goes | 5     |
| Capitalization and punctuation    | WIN-DOW                               | 1     |

|                      |                         |           |
|----------------------|-------------------------|-----------|
| Invalid input        | null, blank, no letters | 3         |
| <b>Total for 3.1</b> |                         | <b>38</b> |

Invalid input throws `IllegalArgumentException` rather than returning a sentinel value, so the three invalid-input tests use `assertThrows`. All 38 passed.

## 2. Screenshot of test execution results

The screenshot below is a current run of the full `SyllableCounterTest` class and reports 61 tests, since the file also holds the 23 sample-word tests added in section 3.2.

```
INFO] Scanning for projects...
INFO]
INFO] -----< edu.cs5588:cs5588-hw1 >-----
INFO] Building cs5588-hw1 1.0-SNAPSHOT
INFO]   from pom.xml
INFO] -----[ jar ]-----
INFO]
INFO] --- resources:3.4.0:resources (default-resources) @ cs5588-hw1 ---
INFO] skip non existing resourceDirectory /Users/joshexe/CS 5588 HW1/src/main/resources
INFO]
INFO] --- compiler:3.15.0:compile (default-compile) @ cs5588-hw1 ---
INFO] Nothing to compile - all classes are up to date.
INFO]
INFO] --- resources:3.4.0:testResources (default-testResources) @ cs5588-hw1 ---
INFO] skip non existing resourceDirectory /Users/joshexe/CS 5588 HW1/src/test/resources
INFO]
INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ cs5588-hw1 ---
INFO] Nothing to compile - all classes are up to date.
INFO]
INFO] --- surefire:3.2.5:test (default-test) @ cs5588-hw1 ---
INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
INFO]
INFO]
INFO] T E S T S
INFO]
INFO] Running SyllableCounterTest
INFO] Tests run: 61, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.132 s -- in SyllableCounterTest
INFO]
INFO] Results:
INFO]
INFO] Tests run: 61, Failures: 0, Errors: 0, Skipped: 0
INFO]
INFO]
INFO] BUILD SUCCESS
INFO]
INFO]
INFO] Total time: 0.954 s
```

## 3. Discussion of AI-generated errors/fixes

- a. The first implementation counted vowel groups and subtracted a silent final e, and it was wrong on common words: whale and smile came back as 2, jumped and makes as 2, and create, poem, and being as 1. I added rules for silent -ed and -es endings, restricted the -le exception to a consonant plus -le, and added patterns for adjacent vowels that are spoken separately. Each of those rules has its own test.

## 4. Source Code: <https://github.com/JUSH334/CS-5588-Homework-1.git>

## 3.2

### 1. Prompts/Response

#### a. Prompt:

- i. Translate the following examples into JUnit tests. Fix all errors and test failures. If any provided syllable counts are incorrect, update them with the correct number and justify your answer. Note that some words can vary slightly in syllable count depending on pronunciation or dialect. Your program will be considered reliable if its results are consistent with those from the syllable counter at <https://syllablecounter.net/count>

#### b. Response:

Translated all 23 supplied examples into a single `@ParameterizedTest` driven by `@CsvSource`, one case per word, and checked every count against the reference counter at [syllablecounter.net](https://syllablecounter.net).

**One provided count was incorrect.** California is listed as 4. The word divides as Ca-li-for-ni-a, with the final *-ia* pronounced as two syllables rather than a single *-ya*. The reference counter returns 5 and gives the division cal-i-for-ni-a, as shown in the screenshot below, so the test asserts 5.

**Eight counts were given as ranges**, so the tests assert the exact value the reference resolves to. Choosing a number inside a stated range is not a correction to the assignment; these are listed for completeness:

| Word                         | Assignment | Division                               | Asserted |
|------------------------------|------------|----------------------------------------|----------|
| Unintentionally              | 6 or more  | un-in-ten-tion-al-ly                   | 6        |
| Misinterpretation            | 6 or more  | mis-in-ter-pre-ta-tion                 | 6        |
| Incomprehensible             | 6 or more  | in-com-pre-hen-si-ble                  | 6        |
| Disproportionately           | 6 or more  | dis-pro-por-tion-ate-ly                | 6        |
| Unconstitutionality          | 7 or more  | un-con-sti-tu-tion-al-i-ties           | 8        |
| Incomprehensibility          | 7 or more  | in-com-pre-hen-si-bil-i-ty             | 8        |
| Internationalization         | 7 or more  | in-ter-na-tion-al-i-za-tion            | 8        |
| Antidisestablishmentarianism | 10 or more | an-ti-dis-es-tab-lish-men-tar-i-an-ism | 11       |

**The remaining 14 counts matched the reference exactly** and were used as given: Sky 1, Rhythm 1, Book 1, Apple 2, Happy 2, Open 2, Animal 3, Chocolate 3, Computer 3, Celebration 4, Voluntary 4, Incomparable 5, Opportunity 5, Misunderstanding 5. Rhythm is counted as 1, the *y* carrying the only vowel sound, though some speakers say rhy-thm.

The 23 sample words are the only tests added in this section; the algorithm's own rules are covered by the 38 tests from 3.1. Every sample word passes from the spelling rules alone, with no word-specific cases in `SyllableCounter`. Full run:

| Test class          | Tests     | Failures | Errors   |
|---------------------|-----------|----------|----------|
| AdjustedAverageTest | 18        | 0        | 0        |
| SyllableCounterTest | 61        | 0        | 0        |
| <b>Total</b>        | <b>79</b> | <b>0</b> | <b>0</b> |

## 2. Screenshot of test execution results

```

[INFO] -----< edit:cs5588:cs5588-hw1 >-----
joshexe@Joshuas-MacBook-Air CS 5588 HW1 % mvn clean test
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO] --- clean:3.2.0:clean (default-clean) @ cs5588-hw1 ---
[INFO] Deleting /Users/joshexe/CS 5588 HW1/target
[INFO] --- resources:3.4.0:resources (default-resources) @ cs5588-hw1 ---
[INFO] skip non existing resourceDirectory /Users/joshexe/CS 5588 HW1/src/main/resources
[INFO] --- compiler:3.15.0:compile (default-compile) @ cs5588-hw1 ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 2 source files with javac [debug release 25] to target/classes
[INFO] --- resources:3.4.0:testResources (default-testResources) @ cs5588-hw1 ---
[INFO] skip non existing resourceDirectory /Users/joshexe/CS 5588 HW1/src/test/resources
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ cs5588-hw1 ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 2 source files with javac [debug release 25] to target/test-classes
[INFO] --- surefire:3.2.5:test (default-test) @ cs5588-hw1 ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] T E S T S
[INFO]
[INFO] Running AdjustedAverageTest
[INFO] Tests run: 18, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.042 s -- in AdjustedAverageTest
[INFO] Running SyllableCounterTest
[INFO] Tests run: 61, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.097 s -- in SyllableCounterTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 79, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] BUILD SUCCESS
[INFO]

```

```

[INFO]
joshexe@Joshuas-MacBook-Air CS 5588 HW1 % for w in california antidisestablishmentarianism; do
  echo "== https://syllablecounter.net/no-of-syllables-in/$w"
  curl -sL -A "Mozilla/5.0" "https://syllablecounter.net/no-of-syllables-in/$w" \
    | sed 's/<[^>]*>/g' | grep -i 'syllables in'
done
== https://syllablecounter.net/no-of-syllables-in/california
  Number of Syllables in the word 'california' - SyllableCounter.net
  Number of syllables in the word 'california'
    No of syllables in california : 5
    Divide syllables in california : cal-i-for-ni-a
== https://syllablecounter.net/no-of-syllables-in/antidisestablishmentarianism
  Number of Syllables in the word 'antidisestablishmentarianism' - SyllableCounter.net
  Number of syllables in the word 'antidisestablishmentarianism'
    No of syllables in antidisestablishmentarianism : 11
    Divide syllables in antidisestablishmentarianism : an-ti-dis-es-tab-lish-men-tar-i-an-ism
joshexe@Joshuas-MacBook-Air CS 5588 HW1 %

```

Image above: reference check against syllablecounter.net for the two counts that needed evidence: California, corrected from the assignment's 4 to 5, and Antidisestablishmentarianism, asserted as 11.



3. Discussion of AI-generated errors/fixes

- a. To make these tests pass, the AI had hardcoded all 23 sample words and their answers in a lookup table inside SyllableCounter, so the tests were checking the table rather than the algorithm. With the table removed, the algorithm was wrong on California (4), Disproportionately (7) and Antidisestablishmentarianism (10). I replaced the table with spelling rules: adjacent vowels spoken separately add a syllable, which covers the "ia" in Ca-li-for-ni-a and an-ti-dis-es-tab-lish-men-tar-i-an-ism, and "-ely" removes one, which covers dis-pro-por-tion-ate-ly. The counter now matches all 23 reference counts from spelling alone, and agrees with the reference on roughly 60 additional common words used while testing.
- b. English spelling does not fully determine pronunciation, so a rule-based counter cannot be exact. The counter is still wrong on words like science (returns 1), situation (3) and million (3), and words whose accepted count varies by speaker, such as every, follow the letter-by-letter count. These limitations are documented in the README rather than patched with word-specific special cases, since hiding them in a lookup table is the error this section is about.

4. Source Code: <https://github.com/JUSH334/CS-5588-Homework-1.git>